

Podcast Shorts Factory

Setup Guide. Ten AI agents that turn long podcasts into short-form videos, automatically. This guide takes you from nothing installed to your first finished clip.

Before you start

You need two things installed. Everything else is handled by the installer.

What	Why	Where
Python 3.11+	Runs the agents	python.org/downloads
ffmpeg	Cuts and renders the video	ffmpeg.org/download.html

On Windows you can install both with one command each:

```
winget install -e --id Python.Python.3.11
winget install -e --id Gyan.FFmpeg
```

On macOS: `brew install python@3.11 ffmpeg`. On Debian or Ubuntu: `sudo apt install python3.11 python3.11-venv ffmpeg`.

Important on Windows: when the Python installer opens, tick "Add python.exe to PATH" on the first screen. If you miss it, the installer will not find Python later.

Step 1. Get the code

Either download the ZIP from the GitHub page (green Code button, then Download ZIP) and unzip it somewhere sensible, or clone it:

```
git clone https://github.com/<your-username>/podcast-shorts-factory.git
```

Step 2. Install

- **Windows:** double-click `install.bat`
- **macOS or Linux:** open a terminal in the folder and run `bash install.sh`

The installer checks Python and ffmpeg, creates an isolated environment, installs the dependencies (this takes two to five minutes the first time), and then launches the setup wizard automatically.

Step 3. The setup wizard

The wizard asks a handful of questions and writes all the config for you. There are no files to edit by hand. You can re-run it any time with `.\factory setup`.

Question 1: which AI provider

This is the brain behind the agents. The default is a free option. Pick one, then paste its key when asked.

Provider	Cost	Get a key at
OpenRouter	Free tier	openrouter.ai/keys
Groq	Free	console.groq.com/keys
Google Gemini	Free	aistudio.google.com/apikey
Ollama	Free, runs locally	ollama.com
Anthropic (Claude)	Paid, best quality	console.anthropic.com

Getting a free key takes about two minutes: sign up, click create key, copy it, paste it into the wizard. You can leave it blank and add it later to the `.env` file.

Question 2: your niche

Pick from presets (money and business, health and fitness, mindset, comedy, sports, tech and AI, relationships) or type your own. This steers what the Trend Scout looks for and what the Finder considers clip-worthy.

Question 3: platforms and privacy

Choose which platforms you want. YouTube posts through the official API. TikTok and Instagram are exported to a folder with captions, ready for you or a scheduler to post.

Start with privacy set to **private**. Let it produce a few clips, look at them yourself, and only switch to public once you are happy. This is the single best way to avoid regrets.

Questions 4 and 5: call-to-action and notifications

The call-to-action is the text shown in the last two seconds of each clip, for example FOLLOW FOR MORE. Notifications are optional: if you want alerts on your phone, install the free ntfy app and choose your own secret topic name.

Step 4. Your first clip

Run the whole pipeline on one podcast episode:

```
Windows: .\factory auto "https://www.youtube.com/watch?v=VIDEO_ID"
Mac/Linux: ./factory.sh auto "https://www.youtube.com/watch?v=VIDEO_ID"
```

What happens, in order:

- The **Finder** downloads the episode and transcribes it. On a long podcast the first run takes a while, because the transcription model is downloaded once and then cached.

- It scores the best moments and shows you the candidates.
- You approve the ones you like (press a to approve).
- The **Editor** renders each approved clip into a vertical short.
- The **Finishing Editor** quality-checks every render and blocks anything broken.
- The **Uploader** asks before posting anything.

Finished clips land in the output folder, with a cover image and creative notes for each one. Files for TikTok and Instagram land in ready_to_post.

Step 5. Connect YouTube (optional)

Only needed if you want the system to upload for you. You create your own Google credentials, so nothing goes through anyone else.

- Follow `docs/youtube-setup.md` to create an OAuth client in the Google Cloud Console and download the `client_secret.json` file into the project folder.
- Run `.\factory auth-youtube` once. A browser opens, you sign in, and the token is cached locally.

Your credentials never leave your machine. They are also listed in `.gitignore`, so they can never be committed by accident.

Step 6. Go hands-free (optional)

Open `config.yaml` and add the channels or playlists you want to pull from, under `scheduler.sources`. Then:

```
.\factory daily
```

That single command scouts current trends, picks the newest video it has not processed yet, finds and renders the best clips, QA-checks them, and schedules the day's posts. On Windows you can register it to run every morning with `tools\schedule_daily.ps1`.

Command reference

Command	What it does
<code>setup</code>	Re-run the setup wizard
<code>find <url></code>	Download, transcribe and score clip candidates
<code>review</code>	Approve or reject candidates
<code>edit</code>	Render approved clips
<code>finish</code>	QA-review the renders
<code>upload</code>	Post or export to platforms
<code>compile</code>	Build the weekly long-form episode
<code>stats</code>	Refresh metrics and update what it has learned
<code>scout</code>	Refresh current trends
<code>skills</code>	List the installed skill playbooks
<code>auto <url></code>	The whole pipeline in one command
<code>daily</code>	Fully unattended run

Troubleshooting

"Sign in to confirm you are not a bot"

YouTube is blocking an anonymous download. In config.yaml set `finder.cookies_from_browser: chrome` and close Chrome completely first, because it locks its own cookie database. Or simply pick a different video.

ffmpeg not found

It is installed but not on your PATH. Close and reopen the terminal, or reinstall with the winget command above.

Rate limited, or the model returns nothing

Free providers have daily caps. Switch `llm.model` in config.yaml to another free model, or add a second provider as a fallback.

The first run is very slow

Expected. The transcription model downloads once (a few hundred MB) and is cached from then on.

No clips were found

Some episodes genuinely have no strong moments. The system automatically retries with a looser brief. If it still finds nothing, try a more conversational episode.

Please use it responsibly

This automates production, not judgement. Only clip content you have the rights to: your own material, licensed material, fair-use commentary, or content you have permission to use. You are responsible for the platform rules you operate under and for any API costs. Start private, review what it makes, then go public.

This project is free and MIT licensed. It is not perfect, but hopefully with the help of the community we can make it perfect for everyone. Issues and pull requests are very welcome.